

# THE LAPLACE TRANSFORM

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# OUTLINE

- **The Core Concept**
- Definitions (Formulas)
- Common Transform Pairs
- Key Properties (Theorems)
  - Linearity
  - Time Shifting
  - s-Shifting
  - Time Scaling
  - Time Differentiation (First Order)
  - Time Differentiation (n-th Order)
  - Time Integration
  - Convolution
  - s-Differentiation
- Solving Differential Equations

## THE CORE CONCEPT

The Laplace Transform is a powerful integral transform used to solve linear ordinary differential equations (ODEs) and systems of ODEs by converting them into algebraic equations.

- **Purpose:** To convert functions from the **time domain** ( $t$ ) to the **complex frequency domain** ( $s$ ). This transformation simplifies the process of solving differential equations, analyzing linear systems, and studying control systems.
- **Big Picture:** Operations like differentiation and integration in the time domain become simple algebraic operations (like multiplication and division) in the  $s$ -domain. After solving the algebraic problem in the  $s$ -domain, the **Inverse Laplace Transform** is used to convert the solution back to the time domain.

## DEFINITIONS (FORMULAS)

### Definitions & Formulas

#### a) Bilateral (Two-sided) Laplace Transform

(Less common, used for non-causal signals)

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st} dt$$

#### b) Unilateral (One-sided) Laplace Transform

(This is the standard form taught first and used for causal systems)

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt$$

## DEFINITIONS (FORMULAS)

### c) Inverse Laplace Transform

$$f(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds$$

In practice, this complex integral is rarely used directly. Instead, we use **partial fraction expansion** and a table of known transform pairs to find the inverse.

### d) Region of Convergence (ROC)

- The set of complex numbers  $s$  for which the Laplace integral converges.
- **Crucial Insight:** A Laplace Transform is only unique when its ROC is specified. The same  $F(s)$  with different ROCs corresponds to different time-domain signals.

# DEFINITIONS (FORMULAS)

## The Relationship Between the Laplace and Fourier Transforms

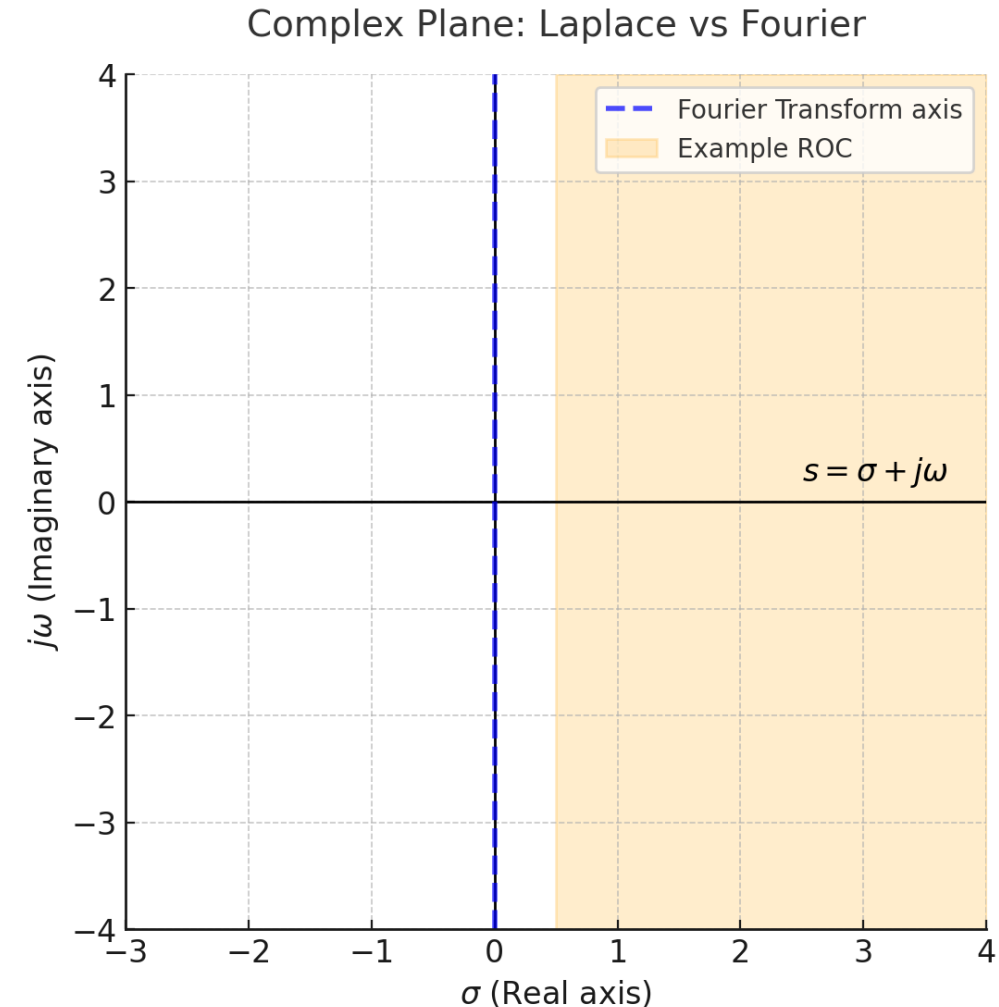
By comparing the two definitions, we arrive at the key relationship:

The Fourier Transform is essentially the Laplace Transform evaluated on the imaginary axis of the s-plane, provided that the axis lies within the Region of Convergence (ROC).

Mathematically, this is expressed as:

$$X(\omega) = F(s) \Big|_{s=j\omega} = \mathcal{L}\{f(t)\} \Big|_{s=j\omega}$$

...but **only if** the ROC of  $F(s)$  includes the imaginary axis ( $\sigma = 0$ ).



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## COMMON TRANSFORM PAIRS

| Transform Pair | Time Domain $f(t)$ | Laplace Domain $F(s)$           | ROC                 |
|----------------|--------------------|---------------------------------|---------------------|
| Unit Impulse   | $\delta(t)$        | 1                               | All $s$             |
| Unit Step      | $u(t)$             | $\frac{1}{s}$                   | $\text{Re}(s) > 0$  |
| Ramp           | $t$                | $\frac{1}{s^2}$                 | $\text{Re}(s) > 0$  |
| Exponential    | $e^{-at}$          | $\frac{1}{s + a}$               | $\text{Re}(s) > -a$ |
| Sine           | $\sin(\omega t)$   | $\frac{\omega}{s^2 + \omega^2}$ | $\text{Re}(s) > 0$  |
| Cosine         | $\cos(\omega t)$   | $\frac{s}{s^2 + \omega^2}$      | $\text{Re}(s) > 0$  |
| n-th Power     | $t^n$              | $\frac{n!}{s^{n+1}}$            | $\text{Re}(s) > 0$  |



# COMMON TRANSFORM PAIRS (UNIT IMPULSE)

## 1. Unit Impulse: $\delta(t) \rightarrow 1$

$$\mathcal{L}\{\delta(t)\} = \int_{0^-}^{\infty} \delta(t)e^{-st}dt = e^{-s \cdot 0} = 1$$



# COMMON TRANSFORM PAIRS (UNIT STEP)

## 2. Unit Step: $u(t) \rightarrow 1/s$

$$\mathcal{L}\{u(t)\} = \int_0^{\infty} e^{-st} dt = \left[ -\frac{e^{-st}}{s} \right]_0^{\infty} = \frac{1}{s}$$



## COMMON TRANSFORM PAIRS (RAMP)

### 3. Ramp: $t \rightarrow 1/s^2$

$$\mathcal{L}\{t\} = \int_0^{\infty} te^{-st} dt = \left[ -\frac{te^{-st}}{s} \right]_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} dt = \frac{1}{s^2}$$

Apply L'Hôpital's Rule:  $\lim_{t \rightarrow \infty} \frac{t}{e^{st}} = \lim_{t \rightarrow \infty} \frac{\frac{d}{dt}[t]}{\frac{d}{dt}[e^{st}]} = \lim_{t \rightarrow \infty} \frac{1}{se^{st}}$

**L'Hôpital's Rule** is a method used to evaluate limits when they are in indeterminate forms such as  $0/0$  or  $\infty/\infty$ .



## COMMON TRANSFORM PAIRS (EXPONENTIAL)

### 4. Exponential: $e^{-at} \rightarrow 1/(s+a)$

$$\mathcal{L}\{e^{-at}\} = \int_0^{\infty} e^{-(s+a)t} dt = \left[ -\frac{e^{-(s+a)t}}{s+a} \right]_0^{\infty} = \frac{1}{s+a}$$



## COMMON TRANSFORM PAIRS (SINE)

5. Sine:  $\sin(\omega t) \rightarrow \omega/(s^2 + \omega^2)$

$$\begin{aligned}\mathcal{L}\{\sin(\omega t)\} &= \int_0^{\infty} \frac{e^{j\omega t} - e^{-j\omega t}}{2j} e^{-st} dt \\ &= \frac{1}{2j} \left( \frac{1}{s - j\omega} - \frac{1}{s + j\omega} \right) = \frac{\omega}{s^2 + \omega^2}\end{aligned}$$



## COMMON TRANSFORM PAIRS (COSINE)

**6. Cosine:  $\cos(\omega t) \rightarrow s/(s^2 + \omega^2)$**

$$\begin{aligned}\mathcal{L}\{\cos(\omega t)\} &= \int_0^\infty \frac{e^{j\omega t} + e^{-j\omega t}}{2} e^{-st} dt \\ &= \frac{1}{2} \left( \frac{1}{s - j\omega} + \frac{1}{s + j\omega} \right) = \frac{s}{s^2 + \omega^2}\end{aligned}$$



## COMMON TRANSFORM PAIRS (N<sup>TH</sup> POWER)

### 7. n-th Power: $t^n \rightarrow n!/s^{n+1}$

$$\mathcal{L}\{t^n\} = \int_0^\infty t^n e^{-st} dt$$

$$= \frac{n}{s} \mathcal{L}\{t^{n-1}\} = \frac{n!}{s^n} \mathcal{L}\{1\} = \frac{n!}{s^{n+1}}$$



## COMMON TRANSFORM PAIRS (N<sup>TH</sup> POWER)

**Integration by parts:**

Let  $u = t^n$ ,  $dv = e^{-st} dt$

Then  $du = nt^{n-1} dt$ ,  $v = -\frac{e^{-st}}{s}$

$$\int_0^{\infty} t^n e^{-st} dt = \left[ -\frac{t^n e^{-st}}{s} \right]_0^{\infty} + \frac{n}{s} \int_0^{\infty} t^{n-1} e^{-st} dt$$

**Evaluate boundary term:**

$$\lim_{t \rightarrow \infty} -\frac{t^n e^{-st}}{s} = 0 \quad (\text{for } \operatorname{Re}(s) > 0)$$

$$\left[ -\frac{t^n e^{-st}}{s} \right]_0^{\infty} = 0 - 0 = 0$$

**Recursive relation:**

$$\mathcal{L}\{t^n\} = \frac{n}{s} \mathcal{L}\{t^{n-1}\}$$



## COMMON TRANSFORM PAIRS (N<sup>TH</sup> POWER)

Apply recursion:

$$\mathcal{L}\{t^n\} = \frac{n}{s} \cdot \frac{n-1}{s} \mathcal{L}\{t^{n-2}\} = \frac{n(n-1)}{s^2} \mathcal{L}\{t^{n-2}\}$$

Continue pattern:

$$\mathcal{L}\{t^n\} = \frac{n(n-1)(n-2) \cdots 1}{s^n} \mathcal{L}\{t^0\}$$

Final result:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^n} \mathcal{L}\{1\} = \frac{n!}{s^n} \cdot \frac{1}{s} = \frac{n!}{s^{n+1}}$$



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## KEY PROPERTIES (THEOREMS)

| Property                | Time Domain             | s-Domain                                            |
|-------------------------|-------------------------|-----------------------------------------------------|
| Linearity               | $af(t) + bg(t)$         | $aF(s) + bG(s)$                                     |
| Time Shifting           | $f(t - a)u(t - a)$      | $e^{-as}F(s)$                                       |
| s-Shifting              | $e^{-at}f(t)$           | $F(s + a)$                                          |
| Time Scaling            | $f(at)$                 | $\frac{1}{a}F\left(\frac{s}{a}\right)$              |
| Differentiation in Time | $\frac{df}{dt}$         | $sF(s) - f(0^-)$                                    |
| Differentiation in Time | $\frac{d^n f}{dt^n}$    | $s^n F(s) - s^{n-1}f(0^-) - \dots - f^{(n-1)}(0^-)$ |
| Integration in Time     | $\int_0^t f(\tau)d\tau$ | $\frac{1}{s}F(s)$                                   |
| Convolution in Time     | $f(t) * g(t)$           | $F(s) \cdot G(s)$                                   |
| Differentiation in s    | $-tf(t)$                | $\frac{d}{ds}F(s)$                                  |



# KEY PROPERTIES (LINEARITY)

**1. Linearity:**  $af(t) + bg(t) \rightarrow aF(s) + bG(s)$

$$\begin{aligned}\mathcal{L}\{af(t) + bg(t)\} &= \int_0^{\infty} [af(t) + bg(t)]e^{-st}dt \\ &= a \int_0^{\infty} f(t)e^{-st}dt + b \int_0^{\infty} g(t)e^{-st}dt = aF(s) + bG(s)\end{aligned}$$



# KEY PROPERTIES (LINEARITY)

## 1. Linearity Example

Find  $\mathcal{L}\{3t + 2e^{-t}\}$

$$\mathcal{L}\{3t + 2e^{-t}\} = 3\mathcal{L}\{t\} + 2\mathcal{L}\{e^{-t}\}$$

$$= 3 \cdot \frac{1}{s^2} + 2 \cdot \frac{1}{s+1} = \frac{3}{s^2} + \frac{2}{s+1}$$



## KEY PROPERTIES (TIME SHIFTING)

**2. Time Shifting:**  $f(t - a)u(t - a) \rightarrow e^{-as}F(s)$

$$\begin{aligned}\mathcal{L}\{f(t - a)u(t - a)\} &= \int_0^{\infty} f(t - a)u(t - a)e^{-st}dt \\ &= \int_a^{\infty} f(t - a)e^{-st}dt = e^{-as} \int_0^{\infty} f(\tau)e^{-s\tau}d\tau = e^{-as}F(s)\end{aligned}$$



# KEY PROPERTIES (TIME SHIFTING)

## 2. Time Shifting Example

Find  $\mathcal{L}\{(t - 2)^2 u(t - 2)\}$

$$\mathcal{L}\{t^2\} = \frac{2}{s^3}$$

$$\mathcal{L}\{(t - 2)^2 u(t - 2)\} = e^{-2s} \cdot \frac{2}{s^3}$$



## KEY PROPERTIES (S-SHIFTING)

**3. s-Shifting:**  $e^{-at}f(t) \rightarrow F(s + a)$

$$\mathcal{L}\{e^{-at}f(t)\} = \int_0^{\infty} e^{-at}f(t)e^{-st}dt$$

$$= \int_0^{\infty} f(t)e^{-(s+a)t}dt = F(s + a)$$



# KEY PROPERTIES (S-SHIFTING)

## 3. s-Shifting Example

Find  $\mathcal{L}\{e^{-3t}t^2\}$

$$\mathcal{L}\{t^2\} = \frac{2}{s^3}$$

$$\mathcal{L}\{e^{-3t}t^2\} = \frac{2}{(s+3)^3}$$



## KEY PROPERTIES (TIME SCALING)

**4. Time Scaling:**  $f(at) \rightarrow \frac{1}{a}F\left(\frac{s}{a}\right)$

$$\begin{aligned}\mathcal{L}\{f(at)\} &= \int_0^{\infty} f(at)e^{-st}dt \\ &= \frac{1}{a} \int_0^{\infty} f(\tau)e^{-(s/a)\tau}d\tau = \frac{1}{a}F\left(\frac{s}{a}\right)\end{aligned}$$



# KEY PROPERTIES (TIME SCALING)

## 4. Time Scaling Example

Find  $\mathcal{L}\{f(4t)\}$  where  $f(t) = e^{-t}$

$$\mathcal{L}\{e^{-t}\} = \frac{1}{s+1}$$

$$\mathcal{L}\{f(4t)\} = \frac{1}{4} \cdot \frac{1}{\frac{s}{4} + 1} = \frac{1}{s+4}$$



## KEY PROPERTIES (TIME DIFFERENTIATION FIRST ORDER)

5. Time Differentiation (First Order):  $\frac{df}{dt} \rightarrow sF(s) - f(0^-)$

$$\mathcal{L} \left\{ \frac{df}{dt} \right\} = \int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt$$

Using integration by parts:

Let  $u = e^{-st}$ ,  $dv = \frac{df}{dt} dt$

Then  $du = -se^{-st} dt$ ,  $v = f(t)$

$$\begin{aligned} \int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt &= [f(t)e^{-st}]_{0^-}^{\infty} - \int_{0^-}^{\infty} f(t)(-se^{-st}) dt \\ &= \left[ \lim_{t \rightarrow \infty} f(t)e^{-st} - f(0^-)e^0 \right] + s \int_{0^-}^{\infty} f(t)e^{-st} dt \end{aligned}$$

Assuming  $\lim_{t \rightarrow \infty} f(t)e^{-st} = 0$  for  $\text{Re}(s)$  large enough:

$$= -f(0^-) + sF(s)$$



# KEY PROPERTIES (TIME DIFFERENTIATION FIRST ORDER)

## 5. Time Differentiation (First Order) Example

Find  $\mathcal{L}\{\cos(2t)\}$  using  $f(t) = \frac{1}{2} \sin(2t)$

$$f'(t) = \cos(2t), \quad f(0) = 0$$

$$\mathcal{L}\{\cos(2t)\} = s\mathcal{L}\left\{\frac{1}{2} \sin(2t)\right\} - 0 = s \cdot \frac{1}{s^2 + 4} = \frac{s}{s^2 + 4}$$



## KEY PROPERTIES (TIME DIFFERENTIATION N<sup>TH</sup> ORDER)

**6. Time Differentiation (n-th Order):**  $\frac{d^n f}{dt^n} \rightarrow s^n F(s) - s^{n-1} f(0^-) - \dots - f^{(n-1)}(0^-)$

**Base case (n=1):**  $\mathcal{L} \left\{ \frac{df}{dt} \right\} = sF(s) - f(0^-)$

**Inductive step:**

Assume true for n=k:

$$\mathcal{L} \left\{ \frac{d^k f}{dt^k} \right\} = s^k F(s) - s^{k-1} f(0^-) - s^{k-2} f'(0^-) - \dots - f^{(k-1)}(0^-)$$

$$\begin{aligned} \text{For } n=k+1: \quad \mathcal{L} \left\{ \frac{d^{k+1} f}{dt^{k+1}} \right\} &= \mathcal{L} \left\{ \frac{d}{dt} \left[ \frac{d^k f}{dt^k} \right] \right\} = s \mathcal{L} \left\{ \frac{d^k f}{dt^k} \right\} - \left. \frac{d^k f}{dt^k} \right|_{t=0^-} \\ &= s \left[ s^k F(s) - s^{k-1} f(0^-) - s^{k-2} f'(0^-) - \dots - f^{(k-1)}(0^-) \right] - f^{(k)}(0^-) \\ &= s^{k+1} F(s) - s^k f(0^-) - s^{k-1} f'(0^-) - \dots - s f^{(k-1)}(0^-) - f^{(k)}(0^-) \end{aligned}$$

**General formula:**  $\mathcal{L} \left\{ \frac{d^n f}{dt^n} \right\} = s^n F(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^-)$



# KEY PROPERTIES (TIME DIFFERENTIATION N<sup>TH</sup> ORDER)

## 6. Time Differentiation (n-th Order) Example

Find  $\mathcal{L}\{t^2\}$  using  $f(t) = t^3/6$

$$f''(t) = t, \quad f'(0) = 0, \quad f(0) = 0$$

$$\mathcal{L}\{t\} = s^2 \mathcal{L}\left\{\frac{t^3}{6}\right\} - s \cdot 0 - 0 = \frac{s^2}{s^4} = \frac{1}{s^2}$$



## KEY PROPERTIES (TIME INTEGRATION)

**7. Time Integration:**  $\int_0^t f(\tau)d\tau \rightarrow \frac{1}{s}F(s)$

$$\begin{aligned}\mathcal{L}\left\{\int_0^t f(\tau)d\tau\right\} &= \int_0^\infty \left[\int_0^t f(\tau)d\tau\right]e^{-st}dt \\ &= \left[-\frac{e^{-st}}{s}\int_0^t f(\tau)d\tau\right]_0^\infty + \frac{1}{s}\int_0^\infty f(t)e^{-st}dt = \frac{1}{s}F(s)\end{aligned}$$



# KEY PROPERTIES (TIME INTEGRATION)

## 7. Time Integration Example

Find  $\mathcal{L}\{\int_0^t \tau d\tau\}$

$$\mathcal{L}\{t\} = \frac{1}{s^2}$$

$$\mathcal{L}\{\int_0^t \tau d\tau\} = \frac{1}{s} \cdot \frac{1}{s^2} = \frac{1}{s^3}$$



## KEY PROPERTIES (CONVOLUTION)

**8. Convolution:**  $f(t) * g(t) \rightarrow F(s)G(s)$

$$\begin{aligned}\mathcal{L}\{f(t) * g(t)\} &= \int_0^\infty \left[ \int_0^t f(\tau)g(t-\tau)d\tau \right] e^{-st} dt \\ &= \int_0^\infty f(\tau) \left[ \int_\tau^\infty g(t-\tau)e^{-st} dt \right] d\tau = F(s)G(s)\end{aligned}$$



# KEY PROPERTIES (CONVOLUTION)

## 8. Convolution Example

Find  $\mathcal{L}\left\{\int_0^t \sin(\tau) \cdot e^{-(t-\tau)} d\tau\right\}$

$$\mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1}, \quad \mathcal{L}\{e^{-t}\} = \frac{1}{s + 1}$$

$$\mathcal{L}\{\sin t * e^{-t}\} = \frac{1}{s^2 + 1} \cdot \frac{1}{s + 1}$$



## KEY PROPERTIES (S-DIFFERENTIATION)

**9. s-Differentiation:**  $-tf(t) \rightarrow \frac{d}{ds}F(s)$

$$\begin{aligned}\frac{d}{ds}F(s) &= \frac{d}{ds} \int_0^{\infty} f(t)e^{-st}dt \\ &= \int_0^{\infty} f(t) \frac{d}{ds}[e^{-st}]dt = \int_0^{\infty} [-tf(t)]e^{-st}dt = \mathcal{L}\{-tf(t)\}\end{aligned}$$



# KEY PROPERTIES (S-DIFFERENTIATION)

## 9. s-Differentiation Example

Find  $\mathcal{L}\{t \cos(2t)\}$

$$\mathcal{L}\{\cos(2t)\} = \frac{s}{s^2 + 4}$$

$$\frac{d}{ds} \left( \frac{s}{s^2 + 4} \right) = \frac{(s^2 + 4) - s(2s)}{(s^2 + 4)^2} = \frac{4 - s^2}{(s^2 + 4)^2}$$

$$\mathcal{L}\{t \cos(2t)\} = -\frac{4 - s^2}{(s^2 + 4)^2} = \frac{s^2 - 4}{(s^2 + 4)^2}$$



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- **Solving Differential Equations**

# SOLVING DIFFERENTIAL EQUATIONS

The Laplace Transform is a powerful integral transform used to solve linear ordinary differential equations (ODEs) and systems of ODEs by converting them into algebraic equations.

## Progress :

1. **Transform:** Take the Laplace Transform of both sides of the differential equation, using the differentiation properties to incorporate initial conditions.
2. **Solve Algebraically:** Solve the resulting algebraic equation for  $F(s)$ .
3. **Expand:** Use partial fraction expansion to break  $F(s)$  into simple terms that appear in the transform table.
4. **Invert:** Find the inverse Laplace Transform of each simple term to get the time-domain solution  $f(t)$ .



# SOLVING DIFFERENTIAL EQUATIONS

## Example 1: First-Order Differential Equation

Solve  $y' + 2y = 0$ ,  $y(0) = 1$

$$\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = 0$$

$$sY(s) - 1 + 2Y(s) = 0$$

$$(s + 2)Y(s) = 1$$

$$Y(s) = \frac{1}{s + 2}$$

$$y(t) = e^{-2t}$$



# SOLVING DIFFERENTIAL EQUATIONS

## Example 2: Second-Order Differential Equation

Solve  $y'' + 4y = 0$ ,  $y(0) = 0$ ,  $y'(0) = 2$

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y\} = 0$$

$$s^2Y(s) - sy(0) - y'(0) + 4Y(s) = 0$$

$$s^2Y(s) - 2 + 4Y(s) = 0$$

$$(s^2 + 4)Y(s) = 2$$

$$Y(s) = \frac{2}{s^2 + 4}$$

$$y(t) = \sin(2t)$$



# SOLVING DIFFERENTIAL EQUATIONS

## Example 3: Nonhomogeneous Equation

Solve  $y' + y = e^{-t}$ ,  $y(0) = 0$

$$\mathcal{L}\{y'\} + \mathcal{L}\{y\} = \mathcal{L}\{e^{-t}\}$$

$$sY(s) - 0 + Y(s) = \frac{1}{s+1}$$

$$(s+1)Y(s) = \frac{1}{s+1}$$

$$Y(s) = \frac{1}{(s+1)^2}$$

$$y(t) = te^{-t}$$



# SOLVING DIFFERENTIAL EQUATIONS

## Example 4: System with Repeated Roots and Initial Conditions

Solve  $y'' + 4y' + 4y = 0$ ,  $y(0) = 1$ ,  $y'(0) = 0$

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y'\} + 4\mathcal{L}\{y\} = 0$$

$$[s^2Y(s) - sy(0) - y'(0)] + 4[sY(s) - y(0)] + 4Y(s) = 0$$

$$s^2Y(s) - s + 4sY(s) - 4 + 4Y(s) = 0$$

$$(s^2 + 4s + 4)Y(s) = s + 4$$

$$Y(s) = \frac{s + 4}{(s + 2)^2} = \frac{s + 2}{(s + 2)^2} + \frac{2}{(s + 2)^2}$$

$$Y(s) = \frac{1}{s + 2} + \frac{2}{(s + 2)^2}$$

$$y(t) = e^{-2t} + 2te^{-2t}$$



# SOLVING DIFFERENTIAL EQUATIONS

## Example 7: System with Exponential Forcing and Initial Conditions

Solve  $y'' - 3y' + 2y = e^{3t}$ ,  $y(0) = 0$ ,  $y'(0) = 1$

$$\mathcal{L}\{y''\} - 3\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = \mathcal{L}\{e^{3t}\}$$

$$[s^2Y(s) - s \cdot 0 - 1] - 3[sY(s) - 0] + 2Y(s) = \frac{1}{s - 3}$$

$$s^2Y(s) - 1 - 3sY(s) + 2Y(s) = \frac{1}{s - 3}$$

$$(s^2 - 3s + 2)Y(s) = 1 + \frac{1}{s - 3}$$

$$Y(s) = \boxed{\frac{1}{s^2 - 3s + 2}} + \boxed{\frac{1}{(s - 3)(s^2 - 3s + 2)}}$$



# SOLVING DIFFERENTIAL EQUATIONS

$$Y(s) = \frac{1}{s^2 - 3s + 2} + \frac{1}{(s - 3)(s^2 - 3s + 2)}$$

$$s^2 - 3s + 2 = (s - 1)(s - 2)$$

$$\frac{1}{(s - 1)(s - 2)} = \frac{A}{s - 1} + \frac{B}{s - 2}$$

$$1 = A(s - 2) + B(s - 1)$$

$$A = -1, \quad B = 1$$

$$\frac{1}{(s - 3)(s - 1)(s - 2)} = \frac{C}{s - 3} + \frac{D}{s - 1} + \frac{E}{s - 2}$$

$$1 = C(s - 1)(s - 2) + D(s - 3)(s - 2) + E(s - 3)(s - 1)$$

$$C = \frac{1}{2}, \quad D = -\frac{1}{2}, \quad E = 1$$



# SOLVING DIFFERENTIAL EQUATIONS

$$Y(s) = \left[ -\frac{1}{s-1} + \frac{1}{s-2} \right] + \left[ \frac{1}{2} \cdot \frac{1}{s-3} - \frac{1}{2} \cdot \frac{1}{s-1} + \frac{1}{s-2} \right]$$

$$Y(s) = -\frac{3}{2} \cdot \frac{1}{s-1} + 2 \cdot \frac{1}{s-2} + \frac{1}{2} \cdot \frac{1}{s-3}$$

$$y(t) = -\frac{3}{2}e^t + 2e^{2t} + \frac{1}{2}e^{3t}$$



# APPENDIX

## Integration by Parts: Simple Guide

### 1. The Core Formula

The formula for Integration by Parts is:

$$\int u dv = uv - \int v du$$

This formula provides a way to transform the integral of a product of functions into another, hopefully simpler, integral.

# APPENDIX

## 2. Derivation (From the Product Rule)

The formula is derived directly from the **Product Rule** for differentiation.

$$\frac{d}{dx}[u(x)v(x)] = u(x)\frac{dv}{dx} + v(x)\frac{du}{dx}$$

$$\int \frac{d}{dx}[u(x)v(x)]dx = \int u(x)\frac{dv}{dx}dx + \int v(x)\frac{du}{dx}dx$$

$$u(x)v(x) = \int u(x)dv + \int v(x)du$$

$$\int u(x)dv = u(x)v(x) - \int v(x)du$$